

Tutorial: Custom YOLO Object Detection with RoboFlow

In this tutorial, you'll learn how to create a custom YOLO object detection model using your own classes and annotations with RoboFlow.

Step 1: Install Required Dependencies

```
%pip install --upgrade ultralytics
%pip install opencv-python
%pip install roboflow
%pip install matplotlib
%pip install numpy==1.26.4 --force-reinstall
```

Requirement already satisfied: ultralytics in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (8.4.26)

Requirement already satisfied: numpy>=1.23.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (1.26.4)

Requirement already satisfied: matplotlib>=3.3.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (3.10.8)

Requirement already satisfied: opencv-python>=4.6.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (4.13.0.92)

Requirement already satisfied: pillow>=7.1.2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (12.1.1)

Requirement already satisfied: pyyaml>=5.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (6.0.3)

Requirement already satisfied: requests>=2.23.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (2.32.5)

Requirement already satisfied: scipy>=1.4.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (1.17.1)

Requirement already satisfied: torch>=1.8.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (2.11.0)

Requirement already satisfied: torchvision>=0.9.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (0.26.0)

Requirement already satisfied: psutil>=5.8.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from ultralytics) (7.2.2)

Requirement already satisfied: polars>=0.20.0 in c:\Users\cyanezru\

AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (1.39.3)
Requirement already satisfied: ultralytics-thop>=2.0.18 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from ultralytics) (2.0.18)
Requirement already satisfied: contourpy>=1.0.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (1.5.0)
Requirement already satisfied: packaging>=20.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib>=3.3.0->ultralytics) (26.0)
Requirement already satisfied: pyparsing>=3 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib>=3.3.0->ultralytics) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0.post0)
Collecting numpy>=1.23.0 (from ultralytics)
Using cached numpy-2.4.3-cp311-cp311-win_amd64.whl.metadata (6.6 kB)
Requirement already satisfied: polars-runtime-32==1.39.3 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from polars>=0.20.0->ultralytics) (1.39.3)
Requirement already satisfied: six>=1.5 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from requests>=2.23.0->ultralytics) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from requests>=2.23.0->ultralytics) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from requests>=2.23.0->ultralytics) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from requests>=2.23.0->ultralytics) (2026.2.25)
Requirement already satisfied: filelock in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (3.25.2)

Requirement already satisfied: typing-extensions>=4.10.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from torch>=1.8.0->ultralytics) (4.15.0)

Requirement already satisfied: setuptools<82 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (65.5.0)

Requirement already satisfied: sympy>=1.13.3 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (3.6.1)

Requirement already satisfied: jinja2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from torch>=1.8.0->ultralytics) (2026.2.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.3)

Using cached numpy-2.4.3-cp311-cp311-win_amd64.whl (12.6 MB)

Installing collected packages: numpy

Attempting uninstall: numpy

Found existing installation: numpy 1.26.4

Uninstalling numpy-1.26.4:

Successfully uninstalled numpy-1.26.4

Successfully installed numpy-2.4.3

Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

WARNING: Failed to remove contents in a temporary directory 'C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages\~--mpy.libs'.

You can safely remove it manually.

WARNING: Failed to remove contents in a temporary directory 'C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages\~--mpy'.

You can safely remove it manually.

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

scikit-learn 1.3.2 requires numpy<2.0,>=1.17.3, but you have numpy 2.4.3 which is incompatible.

tensorflow 2.21.0 requires keras>=3.12.0, but you have keras 2.15.0 which is incompatible.
tensorflow 2.21.0 requires ml_dtypes<1.0.0,>=0.5.1, but you have ml_dtypes 0.2.0 which is incompatible.
tensorflow 2.21.0 requires protobuf<8.0.0,>=6.31.1, but you have protobuf 4.25.8 which is incompatible.
WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

Requirement already satisfied: opencv-python in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (4.13.0.92)
Requirement already satisfied: numpy>=2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from opencv-python) (2.4.3)
Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)
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Requirement already satisfied: roboflow in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (1.2.16)
Requirement already satisfied: certifi in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2026.2.25)
Requirement already satisfied: idna==3.7 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (3.7)
Requirement already satisfied: cyciler in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (0.12.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.5.0)
Requirement already satisfied: matplotlib in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (3.10.8)
Requirement already satisfied: numpy>=1.18.5 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2.4.3)
Requirement already satisfied: opencv-python-headless==4.10.0.84 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (4.10.0.84)
Requirement already satisfied: Pillow>=7.1.2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (12.1.1)
Requirement already satisfied: pillow-avif-plugin<2 in c:\Users\

cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages
(from roboflow) (1.5.5)
Requirement already satisfied: python-dateutil in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from roboflow)
(2.9.0.post0)
Requirement already satisfied: python-dotenv in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.2.2)
Requirement already satisfied: requests in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2.32.5)
Requirement already satisfied: six in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from roboflow) (1.17.0)
Requirement already satisfied: urllib3>=1.26.6 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2.6.3)
Requirement already satisfied: tqdm>=4.41.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (4.67.3)
Requirement already satisfied: PyYAML>=5.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (6.0.3)
Requirement already satisfied: requests-toolbelt in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.0.0)
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Requirement already satisfied: pi-heif<2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.3.0)
Requirement already satisfied: colorama in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from tqdm>=4.41.0->roboflow) (0.4.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (1.3.3)
Requirement already satisfied: fonttools>=4.22.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (4.62.1)
Requirement already satisfied: packaging>=20.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib->roboflow) (26.0)
Requirement already satisfied: pyparsing>=3 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (3.3.2)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages

(from requests->roboflow) (3.4.6)

Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

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WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

Requirement already satisfied: matplotlib in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (3.10.8)

Requirement already satisfied: contourpy>=1.0.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (1.3.3)

Requirement already satisfied: cyciler>=0.10 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (4.62.1)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (1.5.0)

Requirement already satisfied: numpy>=1.23 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (2.4.3)

Requirement already satisfied: packaging>=20.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib) (26.0)

Requirement already satisfied: pillow>=8 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (12.1.1)

Requirement already satisfied: pyparsing>=3 in c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib) (3.3.2)

Requirement already satisfied: python-dateutil>=2.7 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib) (2.9.0.post0)

Requirement already satisfied: six>=1.5 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)

```
Collecting numpy==1.26.4
  Using cached numpy-1.26.4-cp311-cp311-win_amd64.whl.metadata (61 kB)
Using cached numpy-1.26.4-cp311-cp311-win_amd64.whl (15.8 MB)
Installing collected packages: numpy
  Attempting uninstall: numpy
    Found existing installation: numpy 2.4.3
    Uninstalling numpy-2.4.3:
      Successfully uninstalled numpy-2.4.3
Successfully installed numpy-1.26.4
Note: you may need to restart the kernel to use updated packages.
```

```
WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\
AppData\Local\Programs\Python\Python311\Lib\site-packages)
WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\
AppData\Local\Programs\Python\Python311\Lib\site-packages)
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
opencv-python 4.13.0.92 requires numpy>=2; python_version >= "3.9",
but you have numpy 1.26.4 which is incompatible.
tensorflow 2.21.0 requires keras>=3.12.0, but you have keras 2.15.0
which is incompatible.
tensorflow 2.21.0 requires ml_dtypes<1.0.0,>=0.5.1, but you have ml-
dtypes 0.2.0 which is incompatible.
tensorflow 2.21.0 requires protobuf<8.0.0,>=6.31.1, but you have
protobuf 4.25.8 which is incompatible.
WARNING: Ignoring invalid distribution ~atplotlib (c:\Users\cyanezru\
AppData\Local\Programs\Python\Python311\Lib\site-packages)
```

Step 2: Import Required Libraries

```
import cv2
from roboflow import Roboflow
from ultralytics import YOLO
import matplotlib.pyplot as plt
import numpy as np
import os
from IPython.display import Image, display
```

Step 3: Set Up RoboFlow Account

1. Go to RoboFlow and create a free account
2. Create a new project and select "Object Detection" as project type
3. Choose "YOLOv26" as the model format

Step 4: Upload and Annotate Your Images

Using RoboFlow Web Interface:

1. Click "Upload" to add your images

2. Use the annotation tool to draw bounding boxes around objects
3. Assign the correct class labels to each bounding box
4. Apply preprocessing and augmentation as needed

Step 5. Download your Custom your Dataset

1. In RoboFlow, go to your project dashboard
2. Click "Generate" to create a dataset version
3. Apply any desired augmentations
4. Click "Export" and select "YOLOvX" format (x the version you are using)
5. Copy the download code

Initialize RoboFlow (you'll need your API key)

```
!pip install roboflow
```

```
from roboflow import Roboflow
rf = Roboflow(api_key="JveMfT08Up00ztjKycz2")
project = rf.workspace("carlosyanezs-
workspace").project("clase3_grupo3")
version = project.version(1)
dataset = version.download("yolo26")
```

Requirement already satisfied: roboflow in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (1.2.16)

Requirement already satisfied: certifi in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2026.2.25)

Requirement already satisfied: idna==3.7 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (3.7)

Requirement already satisfied: cyciler in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (0.12.1)

Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.5.0)

Requirement already satisfied: matplotlib in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (3.10.8)

Requirement already satisfied: numpy>=1.18.5 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.26.4)

Requirement already satisfied: opencv-python-headless==4.10.0.84 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (4.10.0.84)

Requirement already satisfied: Pillow>=7.1.2 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (12.1.1)

Requirement already satisfied: pillow-avif-plugin<2 in C:\Users\

cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages
(from roboflow) (1.5.5)
Requirement already satisfied: python-dateutil in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from roboflow)
(2.9.0.post0)
Requirement already satisfied: python-dotenv in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.2.2)
Requirement already satisfied: requests in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow)
(2.32.5)
Requirement already satisfied: six in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from roboflow) (1.17.0)
Requirement already satisfied: urllib3>=1.26.6 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (2.6.3)
Requirement already satisfied: tqdm>=4.41.0 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (4.67.3)
Requirement already satisfied: PyYAML>=5.3.1 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (6.0.3)
Requirement already satisfied: requests-toolbelt in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.0.0)
Requirement already satisfied: filetype in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.2.0)
Requirement already satisfied: pi-heif<2 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from roboflow) (1.3.0)
Requirement already satisfied: colorama in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from tqdm>=4.41.0->roboflow) (0.4.6)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (1.3.3)
Requirement already satisfied: fonttools>=4.22.0 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (4.62.1)
Requirement already satisfied: packaging>=20.0 in C:\Users\cyanezru\AppData\Roaming\Python\Python311\site-packages (from matplotlib->roboflow) (26.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from matplotlib->roboflow) (3.3.2)
Requirement already satisfied: charset_normalizer<4,>=2 in C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages (from requests->roboflow) (3.4.6)

```
WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)
WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)
WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages)
```

```
loading Roboflow workspace...
loading Roboflow project...
```

Step 6: Train Your Custom YOLO Model

```
# Load the YOLO model with custom configuration
model = YOLO('yolo26n.pt') # Using nano version as base

# Train the model on your custom dataset
results = model.train(
    data=f'{dataset.location}/data.yaml', # Path to dataset config
    epochs=20, # Number of training epochs
    imgsz=640, # Image size
    batch=16, # Batch size
    name='custom_yolo_model' # Name for the training run
)
```

```
Ultralytics 8.4.26 Python-3.11.9 torch-2.11.0+cpu CPU (13th Gen Intel Core i7-1355U)
```

```
engine\trainer: agnostic_nms=False, amp=True, angle=1.0,
augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5,
cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5,
compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip,
cos_lr=False, cutmix=0.0, data=c:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\Clase3_Grupo3-1\data.yaml, degrees=0.0, deterministic=True,
device=cpu, dfl=1.5, dnn=False, dropout=0.0, dynamic=False,
embed=None, end2end=None, epochs=20, erasing=0.4, exist_ok=False,
fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None,
half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False,
iou=0.7, keras=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01,
mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolo26n.pt,
momentum=0.937, mosaic=1.0, multi_scale=0.0, name=custom_yolo_model2,
nbs=64, nms=False, opset=None, optimize=False, optimizer=auto,
overlap_mask=True, patience=100, perspective=0.0, plots=True,
pose=12.0, pretrained=True, profile=False, project=None, rect=False,
resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False,
save_crop=False, save_dir=C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\runs\detect\custom_yolo_model2, save_frames=False, save_json=False,
save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0,
show=False, show_boxes=True, show_conf=True, show_labels=True,
simplify=True, single_cls=False, source=None, split=val,
```

```

stream_buffer=False, task=detect, time=None, tracker=botsort.yaml,
translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False,
warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8,
weight_decay=0.0005, workers=8, workspace=None
Overriding model.yaml nc=80 with nc=2

```

	from	n	params	module
arguments				
0	-1	1	464	ultralytics.nn.modules.conv.Conv
[3, 16, 3, 2]				
1	-1	1	4672	ultralytics.nn.modules.conv.Conv
[16, 32, 3, 2]				
2	-1	1	6640	
ultralytics.nn.modules.block.C3k2				[32, 64, 1, False, 0.25]
3	-1	1	36992	ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]				
4	-1	1	26080	
ultralytics.nn.modules.block.C3k2				[64, 128, 1, False, 0.25]
5	-1	1	147712	ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]				
6	-1	1	87040	
ultralytics.nn.modules.block.C3k2				[128, 128, 1, True]
7	-1	1	295424	ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]				
8	-1	1	346112	
ultralytics.nn.modules.block.C3k2				[256, 256, 1, True]
9	-1	1	164608	
ultralytics.nn.modules.block.SPPF				[256, 256, 5, 3, True]
10	-1	1	249728	
ultralytics.nn.modules.block.C2PSA				[256, 256, 1]
11	-1	1	0	
torch.nn.modules.upsampling.Upsample				[None, 2, 'nearest']
12	[-1, 6]	1	0	
ultralytics.nn.modules.conv.Concat				[1]
13	-1	1	119808	
ultralytics.nn.modules.block.C3k2				[384, 128, 1, True]
14	-1	1	0	
torch.nn.modules.upsampling.Upsample				[None, 2, 'nearest']
15	[-1, 4]	1	0	
ultralytics.nn.modules.conv.Concat				[1]

```

16          -1  1    34304
ultralytics.nn.modules.block.C3k2          [256, 64, 1, True]

17          -1  1    36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]

18      [-1, 13]  1        0
ultralytics.nn.modules.conv.Concat          [1]

19          -1  1    95232
ultralytics.nn.modules.block.C3k2          [192, 128, 1, True]

20          -1  1   147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]

21      [-1, 10]  1        0
ultralytics.nn.modules.conv.Concat          [1]

22          -1  1   463104
ultralytics.nn.modules.block.C3k2          [384, 256, 1, True, 0.5,
True]

23      [16, 19, 22]  1    241956
ultralytics.nn.modules.head.Detect          [2, 1, True, [64, 128,
256]]
YOLO26n summary: 260 layers, 2,504,580 parameters, 2,504,580
gradients, 5.8 GFLOPs

```

```

Transferred 606/708 items from pretrained weights
train: Fast image access (ping: 1.30.2 ms, read: 24.47.5 MB/s, size:
43.5 KB)
train: Scanning C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
Clase3_Grupo3-1\train\labels.cache... 27 images, 0 backgrounds, 0
corrupt: 100% _____ 27/27 13.5Kit/s 0.0s
val: Fast image access (ping: 1.20.2 ms, read: 22.84.9 MB/s, size:
38.6 KB)
val: Scanning C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\
Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
Clase3_Grupo3-1\valid\labels.cache... 21 images, 0 backgrounds, 0
corrupt: 100% _____ 21/21 0.0s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and
'momentum=0.937' and determining best 'optimizer', 'lr0' and
'momentum' automatically...
optimizer: AdamW(lr=0.001667, momentum=0.9) with parameter groups 114
weight(decay=0.0), 126 weight(decay=0.0005), 126 bias(decay=0.0)
Plotting labels to C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model2\labels.jpg...
Image sizes 640 train, 640 val
Using 0 dataloader workers
Logging results to C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\

```

Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
 runs\detect\custom_yolo_model2
 Starting training for 20 epochs...

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	23.0s/it	23.0s		
		all	21	21	0.00253
0.051	0.029				0.75

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	6.7s/it	6.7s		
		all	21	21	0.00254
0.0697	0.0398				0.75

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	6.2s/it	6.2s		
		all	21	21	0.00313
0.08	0.0495				0.875

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	5.6s/it	5.6s		
		all	21	21	0.00356
0.0922	0.0588				0.875

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	4.7s/it	4.7s		
		all	21	21	0.00413
0.0967	0.0657				0.875

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	4.5s/it	4.5s		
		all	21	21	0.00463
0.0715	0.0525				0.875

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	4.2s/it	4.2s		

```
all 21 21 0.00511 0.938
0.0621 0.0487
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
1/1 7.6s/it 7.6s
```

```
all 21 21 0.00543 0.938
0.0606 0.0499
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
1/1 5.3s/it 5.3s
```

```
all 21 21 0.00547 0.938
0.084 0.0702
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
1/1 5.3s/it 5.3s
```

```
all 21 21 0.00558 0.938
0.0943 0.0789
```

Closing dataloader mosaic

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
1/1 6.4s/it 6.4s
```

```
all 21 21 0.00561 0.938
0.155 0.127
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
1/1 4.9s/it 4.9s
```

```
all 21 21 0.00559 0.938
0.165 0.135
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
1/1 5.0s/it 5.0s
```

```
all 21 21 0.00558 0.938
0.189 0.152
```

```
Epoch GPU_mem box_loss cls_loss dfl_loss Instances
Size ages Instances Box(P R mAP50 mAP50-95): 100%
1/1 4.1s/it 4.1s
```

0.189	0.149	all	21	21	0.00545	0.938
-------	-------	-----	----	----	---------	-------

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	4.1s/it	4.1s		

0.191	0.147	all	21	21	0.00504	0.899
-------	-------	-----	----	----	---------	-------

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	5.3s/it	5.3s		

0.195	0.149	all	21	21	0.00496	0.899
-------	-------	-----	----	----	---------	-------

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	5.3s/it	5.3s		

0.198	0.152	all	21	21	0.00494	0.899
-------	-------	-----	----	----	---------	-------

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	5.9s/it	5.9s		

0.2	0.152	all	21	21	0.00525	0.938
-----	-------	-----	----	----	---------	-------

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	6.7s/it	6.7s		

0.204	0.156	all	21	21	0.00521	0.938
-------	-------	-----	----	----	---------	-------

Epoch	GPU_mem	box_loss	cls_loss	df_l_loss	Instances
Size					
ages	Instances	Box(P	R	mAP50	mAP50-95): 100%
	1/1	4.2s/it	4.2s		

0.206	0.157	all	21	21	0.00521	0.938
-------	-------	-----	----	----	---------	-------

20 epochs completed in 0.171 hours.

Optimizer stripped from C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\runs\detect\custom_yolo_model2\weights\last.pt, 5.4MB

Optimizer stripped from C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\

```
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model2\weights\best.pt, 5.4MB
```

```
Validating C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\
Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\runs\
detect\custom_yolo_model2\weights\best.pt...
```

```
Ultralytics 8.4.26 Python-3.11.9 torch-2.11.0+cpu CPU (13th Gen Intel
Core i7-1355U)
```

```
YOLO26n summary (fused): 122 layers, 2,375,226 parameters, 0
gradients, 5.2 GFLOPs
```

ages	Instances	Box(P	R	mAP50	mAP50-95):
1/1	3.8s/it	3.8s			100%
	all		21	21	0.00521 0.938
0.207	0.157	oximeter	8	8	0.00144 0.875
0.173	0.135	pencil sharpener	13	13	0.00898 1
0.241	0.18				

```
Speed: 1.7ms preprocess, 148.2ms inference, 0.0ms loss, 0.4ms
postprocess per image
```

```
Results saved to C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model2
```

Step 7: Evaluate Your Model

```
# Evaluate model performance
metrics = model.val() # Validate the model
print(f"mAP50-95: {metrics.box.map}")
print(f"mAP50: {metrics.box.map50}")
```

```
# Show some validation results
!yolo task=detect mode=val
model=runs/detect/custom_yolo_model/weights/best.pt
data={dataset.location}/data.yaml
```

```
Ultralytics 8.4.26 Python-3.11.9 torch-2.11.0+cpu CPU (13th Gen Intel
Core i7-1355U)
```

```
YOLO26n summary (fused): 122 layers, 2,375,226 parameters, 0
gradients, 5.2 GFLOPs
```

```
val: Fast image access (ping: 0.50.0 ms, read: 28.013.0 MB/s, size:
37.1 KB)
```

```
val: Scanning C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\
Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
Clase3_Grupo3-1\valid\labels.cache... 21 images, 0 backgrounds, 0
corrupt: 100%
```

ages	Instances	Box(P	R	mAP50	mAP50-95):
2/2	2.2s/it	4.4s<10.4s			100%
	all		21	21	0.00521 0.938
0.207	0.157				

0.173	0.135	0.135	0.135	0.135	0.135
0.241	0.18	0.18	0.18	0.18	0.18

Speed: 5.6ms preprocess, 180.8ms inference, 0.0ms loss, 0.8ms postprocess per image

Results saved to C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\runs\detect\val2

mAP50-95: 0.15747078186292382

mAP50: 0.20670191243646846

Traceback (most recent call last):

```
File "<frozen runpy>", line 198, in _run_module_as_main
File "<frozen runpy>", line 88, in _run_code
File "C:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Scripts\yolo.exe\__main__.py", line 5, in <module>
File "c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages\ultralytics\cfg\__init__.py", line 907, in entrypoint
check_dict_alignment(full_args_dict, {a: ""})
File "c:\Users\cyanezru\AppData\Local\Programs\Python\Python311\Lib\site-packages\ultralytics\cfg\__init__.py", line 506, in check_dict_alignment
raise SyntaxError(string + CLI_HELP_MSG) from e
SyntaxError: '-' is not a valid YOLO argument.
```

Arguments received: ['yolo', 'task=detect', 'mode=val', 'model=runs/detect/custom_yolo_model/weights/best.pt', 'data=c:\\Users\\cyanezru\\OneDrive', '-', 'NTT', 'DATA', 'EMEAL\\Documentos\\Repositories\\Vision-por-computador-MIA-B---Grupo-3\\Clase_3\\Clase3_Grupo3-1\data.yaml']. Ultralytics 'yolo' commands use the following syntax:

yolo TASK MODE ARGS

Where TASK (optional) is one of ['pose', 'segment', 'detect', 'obb', 'classify']
 MODE (required) is one of ['track', 'benchmark', 'predict', 'export', 'val', 'train']
 ARGS (optional) are any number of custom 'arg=value' pairs like 'imgsz=320' that override defaults.

See all ARGS at <https://docs.ultralytics.com/usage/cfg> or with 'yolo cfg'

1. Train a detection model for 10 epochs with an initial learning_rate of 0.01
 yolo train data=coco8.yaml model=yolo26n.pt epochs=10 lr0=0.01

2. Predict a YouTube video using a pretrained segmentation model at image size 320:

```
yolo predict model=yolo26n-seg.pt
source='https://youtu.be/LNw0DJXcvt4' imgsz=320
```

3. Validate a pretrained detection model at batch-size 1 and image size 640:

```
yolo val model=yolo26n.pt data=coco8.yaml batch=1 imgsz=640
```

4. Export a YOLO26n classification model to ONNX format at image size 224 by 128 (no TASK required)

```
yolo export model=yolo26n-cls.pt format=onnx imgsz=224,128
```

5. Ultralytics solutions usage

```
yolo solutions count or any of ['crop', 'blur', 'workout',
'heatmap', 'isegment', 'visioneye', 'speed', 'queue', 'analytics',
'inference', 'trackzone'] source="path/to/video.mp4"
```

6. Run special commands:

```
yolo help
yolo checks
yolo version
yolo settings
yolo copy-cfg
yolo cfg
yolo solutions help
```

Docs: <https://docs.ultralytics.com>

Solutions: <https://docs.ultralytics.com/solutions/>

Community: <https://community.ultralytics.com>

GitHub: <https://github.com/ultralytics/ultralytics>

Step 8: Test Your Custom Model on an Image

```
# Load your custom trained model
```

```
custom_model = YOLO(r'C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model\weights\best.pt')
```

```
# Test on a sample image
```

```
results = custom_model(r'C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
dataset\oximeter.jpg', conf=0.25)
```

```
# Display results
```

```
%matplotlib inline
```

```
for result in results:
```

```
    img = result.plot() # Plot detections
```

```
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
    plt.figure(figsize=(12, 8))
```

```
    plt.imshow(img_rgb)
```

```
plt.axis('off')  
plt.show()
```

image 1/1 C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\
Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\dataset\
oximeter.jpg: 640x640 (no detections), 404.7ms
Speed: 45.6ms preprocess, 404.7ms inference, 1.9ms postprocess per
image at shape (1, 3, 640, 640)



Step 9: Test Your Custom Model on Video

```
# Notebook-safe video inference (no cv2.imshow / HighGUI required)
from ultralytics import YOLO
import cv2
from IPython.display import Image as IPyImage, display, clear_output
import time

# Cargar modelo
custom_model = YOLO(r'C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\
Documentos\Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\
runs\detect\custom_yolo_model\weights\best.pt')

# Ruta del video
video_path = r'C:\Users\cyanezru\OneDrive - NTT DATA EMEAL\Documentos\
Repositories\Vision-por-computador-MIA-B---Grupo-3\Clase_3\dataset\
test.mp4'

# Abrir video
cap = cv2.VideoCapture(video_path)
if not cap.isOpened():
    raise RuntimeError(f'No se pudo abrir el video: {video_path}')

names = custom_model.names
frame_idx = 0

try:
    while True:
        ret, frame = cap.read()

        if not ret:
            print('Fin del video o error')
            break

        frame_idx += 1

        # Ejecutar detección
        results = custom_model(frame, conf=0.25, verbose=False)
        result = results[0]

        # Imprimir resumen de resultados por frame
        boxes = result.boxes
        n = 0 if boxes is None else len(boxes)
        print(f'Frame {frame_idx}: {n} detecciones')
        if n > 0:
            for i in range(n):
                cls_id = int(boxes.cls[i].item())
                conf = float(boxes.conf[i].item())
                x1, y1, x2, y2 = boxes.xyxy[i].tolist()
                cls_name = names.get(cls_id, str(cls_id)) if
isinstance(names, dict) else names[cls_id]
```

```

        print(f' - {cls_name}: conf={conf:.2f},
box=({x1:.0f}, {y1:.0f}, {x2:.0f}, {y2:.0f})')

    # Dibujar resultados
    annotated_frame = result.plot()

    # Mostrar frame en el notebook (sin ventanas nativas)
    ok, buffer = cv2.imencode('.jpg', annotated_frame)
    if not ok:
        continue

    clear_output(wait=True)
    display(IPyImage(data=buffer.tobytes()))
    time.sleep(0.05) # Controla velocidad de reproducción
except KeyboardInterrupt:
    print('Interrumpido por el usuario')
finally:
    cap.release()
    clear_output(wait=True)
    print('Recursos liberados')

```

Recursos liberados

Activity: Create Your Custom Detector

1. **Create a RoboFlow account** and set up a new project
2. **Upload at least 25 images** of your custom objects (at least 2)
3. **Annotate all images** with bounding boxes and class labels
4. **Train a YOLO model** on your custom dataset
5. **Test your model** on new images and evaluate its performance
6. **Implement real-time detection** with your custom classes

Grading Rubric:

Criteria	Points	Description
1. Dataset Creation	4	Successfully creates and annotates a custom dataset with at least 25 images and multiple classes in RoboFlow.
2. Model Training	1	Properly trains a YOLOv8 model on the custom dataset with appropriate hyperparameters.
3. Model Evaluation	1	Evaluates model performance with appropriate metrics and provides analysis of results.
4. Custom Detection	3	Implements real-time detection with custom classes and displays results correctly. You have to use

Criteria	Points	Description
5. Documentation	1	video from your webcam, no images or saved videos allowed. Provides clear documentation of the process, including dataset details and model performance.

Total: 10 points

Tips for Success:

- **Quality over quantity:** Well-annotated images are more important than many poorly annotated ones
- **Class balance:** Try to have a similar number of examples for each class
- **Variety:** Include different angles, lighting conditions, and backgrounds
- **Augmentation:** Use RoboFlow's augmentation features to expand your dataset